

Experiment 13: Migration of Local Web App to AWS

Aim

To migrate a locally hosted web application to AWS using EC2 for hosting and S3 for storing static content.

This project demonstrates how a locally hosted web application can be migrated to the AWS cloud platform. In this experiment, an EC2 instance was used as a virtual server to host the application. A web server (Apache) was installed and configured on the EC2 instance to make the application accessible over the internet.

To enable secure access and permissions, IAM roles and policies were created and attached to the EC2 instance. AWS CLI was installed to interact with AWS services from the command line. A simple Python script using the boto3 library was created and executed to verify the connection between the EC2 instance and AWS services.

The successful execution of the script and access through the public IP address confirm that the application has been properly deployed and is functioning correctly in the cloud environment

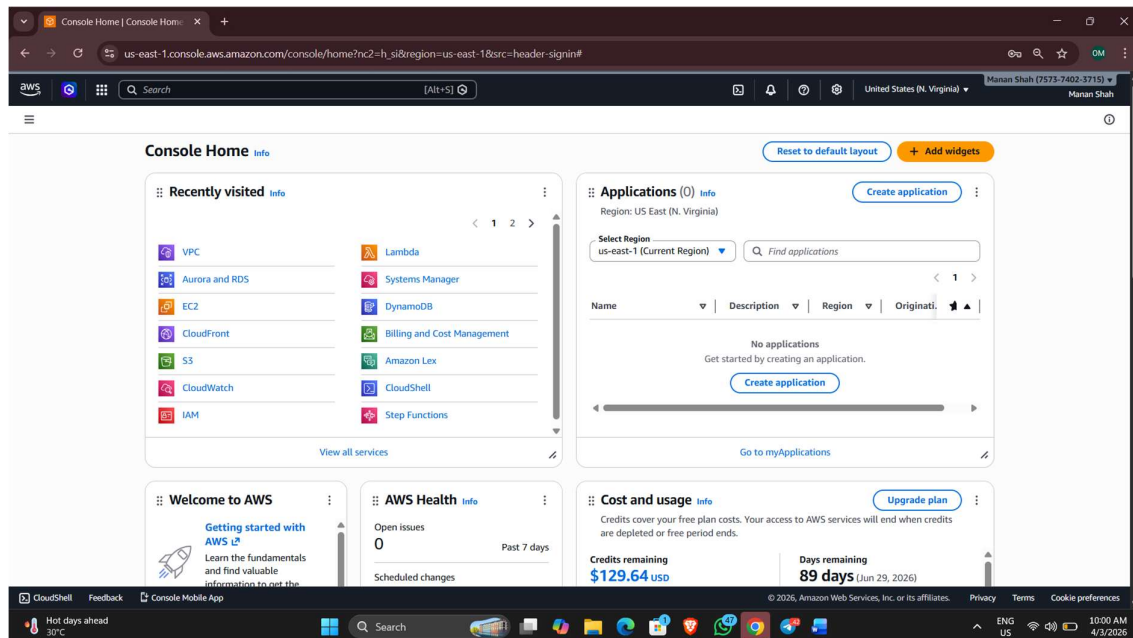
Complete Experiment Flow

1. Logged into AWS account
2. Launched an EC2 instance (virtual server)
3. Connected to EC2 using SSH
4. Installed Apache web server on EC2
5. Started Apache service
6. Accessed the web server using public IP in browser
7. Created an IAM policy for S3 access
8. Created an IAM role and attached the policy
9. Attached the IAM role to EC2 instance
10. Installed AWS CLI on EC2
11. Created a Python script using boto3
12. Installed required Python packages (boto3)
13. Executed the Python script
14. Verified AWS connectivity successfully

Procedure / Steps

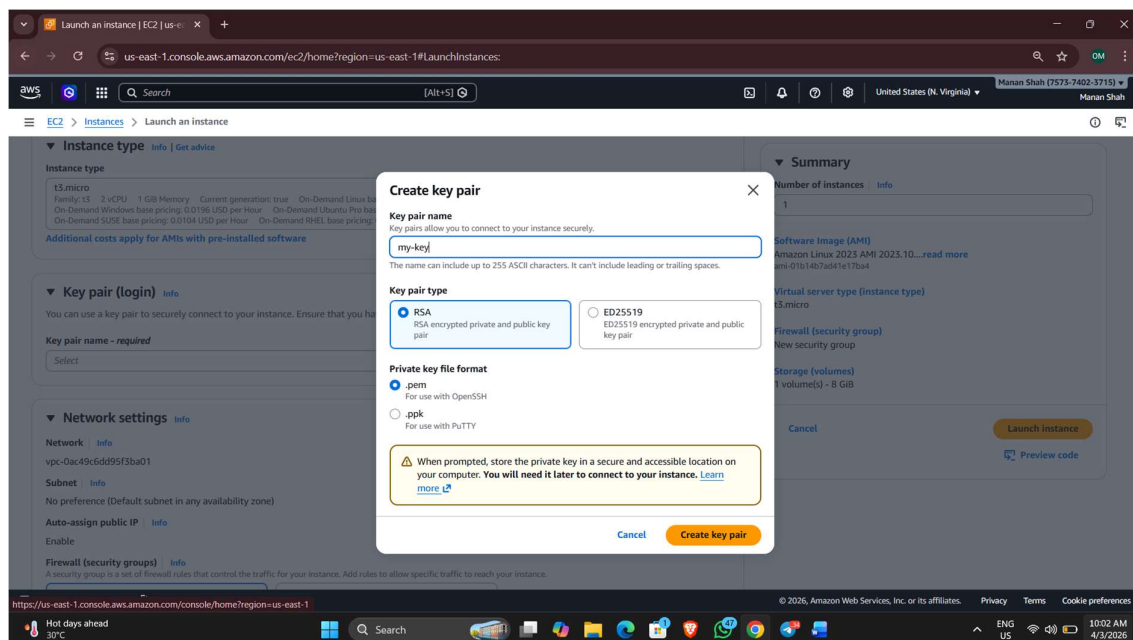
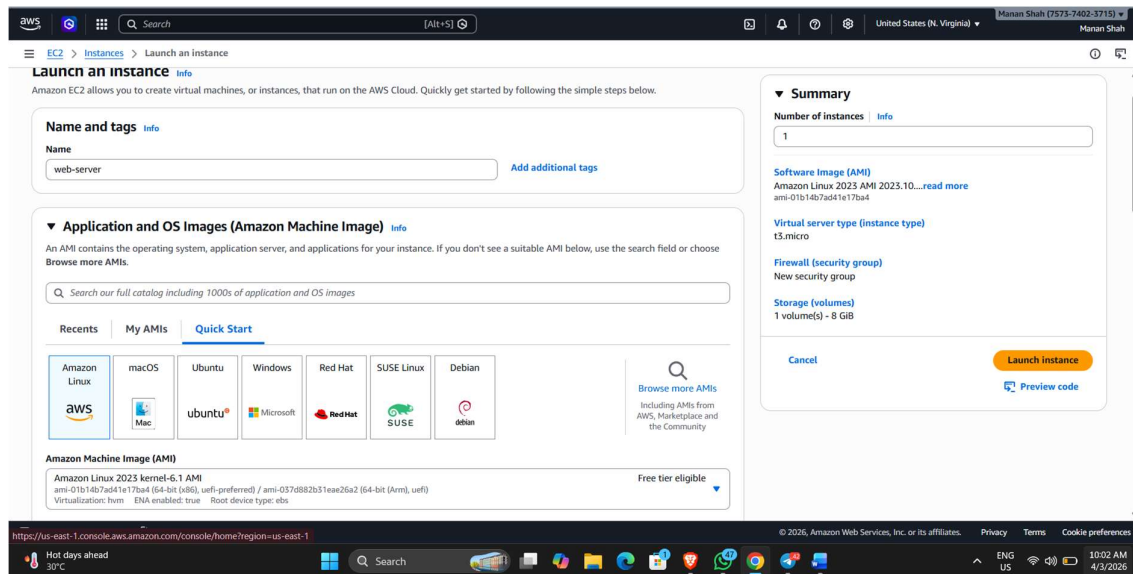
Step 1: Login to AWS

1. Go to AWS website
2. Click on **Sign In**
3. Enter username & password
4. Open **AWS Management Console**



Step 2: Launch EC2 Instance

1. Search **EC2** in services
2. Click **Launch Instance**
3. Choose **Amazon Linux / Ubuntu**
4. Select instance type (t2.micro - free tier)
5. Create or select **Key Pair**
6. Configure security group (Allow HTTP, SSH)
7. Click **Launch Instance**



Than

Network Settings (Security Group)

Click **Edit**

Add rules:

Type Port Source

SSH 22 Anywhere

HTTP 80 Anywhere

Security group name - *required*

launch-wizard-5

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-/[*.@!+=,;()\$%*~]

Description - *required* | Info

launch-wizard-5 created 2026-04-03T04:31:33.516Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type | Info

ssh

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Anywhere

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP: 80, 0.0.0.0/0)

Type | Info

HTTP

Protocol | Info

TCP

Port range | Info

80

Source type | Info

Anywhere

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

Remove

▼ Summary

Number of instances | Info

1

Software image (AMI)

Amazon Linux 2023 AMI 2023.10...read more

ami-01b14b7ad41e17ba4

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

30°C Sunny

Search

ENG US 10:03 AM 4/3/2026

aws

Search [Alt+S]

United States (N. Virginia) Manan Shah (7573-7402-3715) Manan Shah

EC2 > Instances > Launch an instance

Launching instance

Launch initiation

80%

Details

Please wait while we launch your instance.
Do not close your browser while this is loading.

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

30°C Sunny

Search

ENG US 10:04 AM 4/3/2026

aws

Search [Alt+S]

United States (N. Virginia) Manan Shah (7573-7402-3715) Manan Shah

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-0782204a3434aa8a2)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

30°C Sunny

Search

ENG US 10:04 AM 4/3/2026

Step 3: Connect to Instance

1. Go to **Instances**
2. Select your instance
3. Wait 3/3 check pass
4. Click **Connect**

Use command:

```
ssh -i my-key.pem ec2-user@<public-ip>
```

in my case

```
ssh -i my-key.pem ec2-user@3.92.48.227
```

The screenshot displays the AWS Management Console interface for the EC2 Instances page. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of instances. The 'web-server' instance is selected, and the 'Connect' button is highlighted with a red circle. Below the list, the details for the 'web-server' instance are shown, including its ID, state, type, and public IP address.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
mywebserver	i-048a46708f10a482a	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	ec2-52-90-170-66.com...	52.90.1
mywebse	i-00d8b76a37d3acd85	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	ec2-54-91-199-58.com...	54.91.1
web-server	i-0782204a3434aa8a2	Running	t3.micro	Initializing	View alarms +	us-east-1b	ec2-3-92-48-227.comp...	3.92.48.227

i-0782204a3434aa8a2 (web-server)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID: i-0782204a3434aa8a2

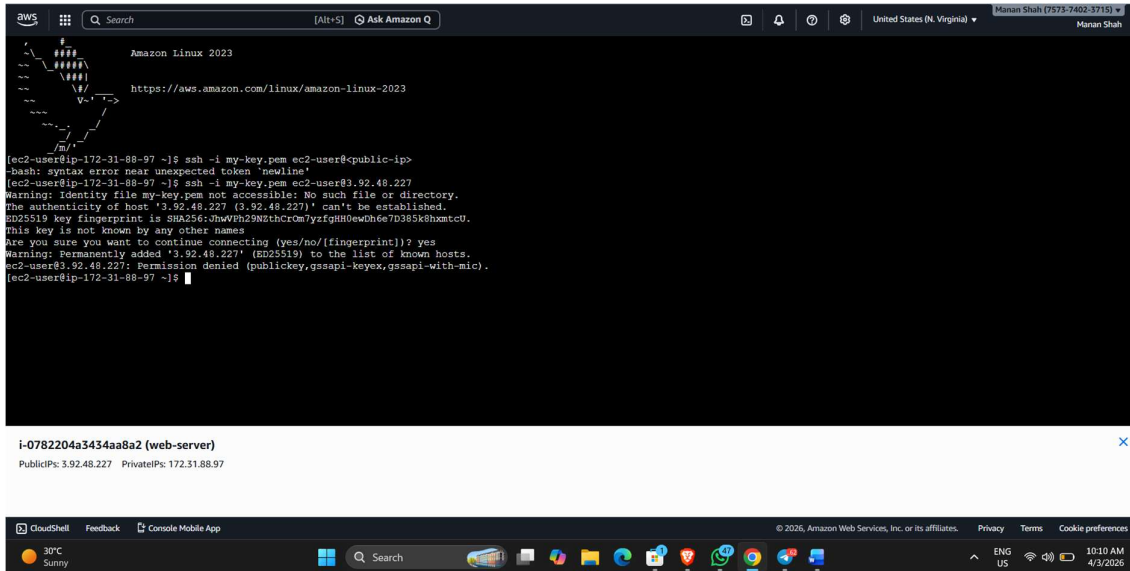
Public IPv4 address: 3.92.48.227 | [open address](#)

Private IPv4 addresses: 172.31.88.97

Instance state: Running

Public DNS: ec2-3-92-48-227.compute-1.amazonaws.com | [open address](#)

Private IP DNS name (IPv4 only): ip-172-31-88-97.ec2.internal

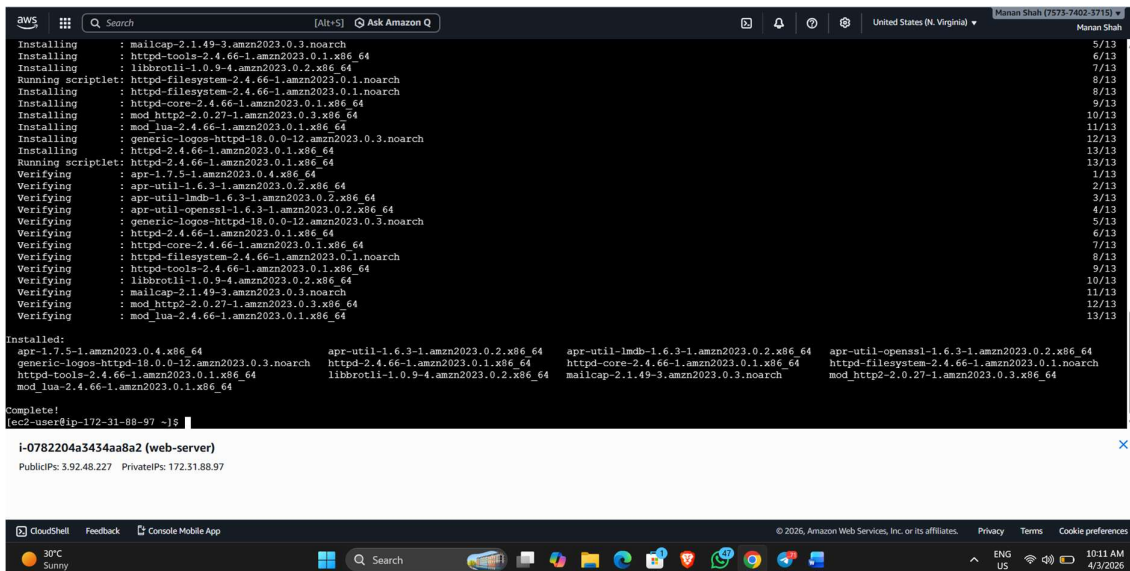


Step 4: Server Setup

sudo yum update -y

sudo yum install httpd -y

sudo systemctl start httpd

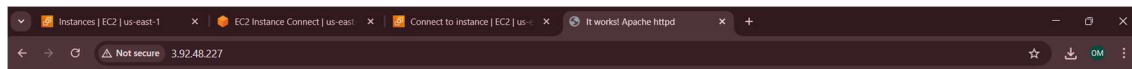


Step 5: Test

Test in Browser

Open browser and enter:

http://3.92.48.227



It works!



Result:

The default Apache web page is displayed.

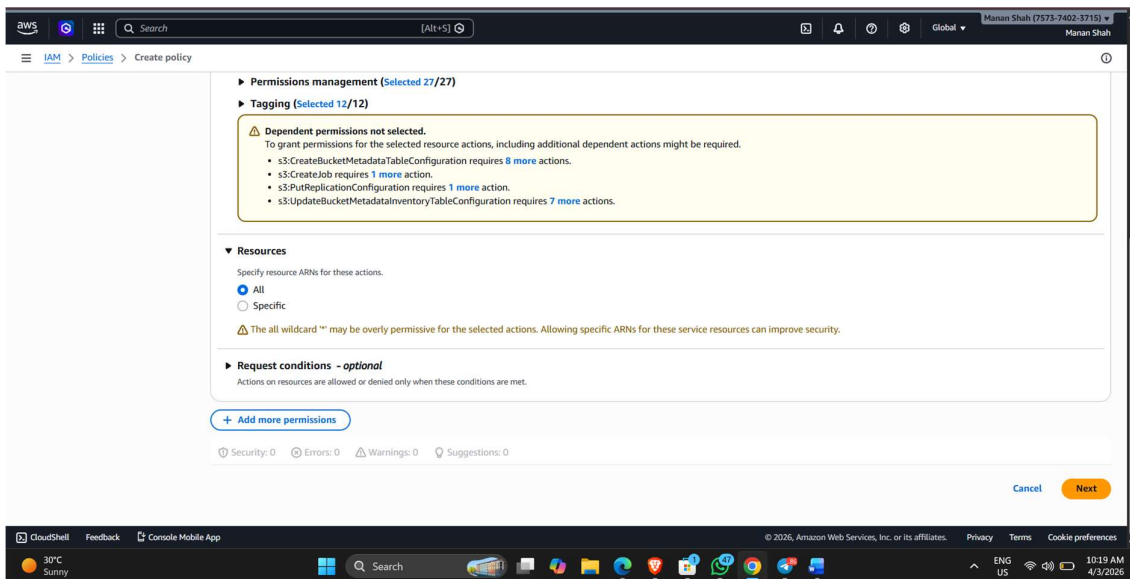
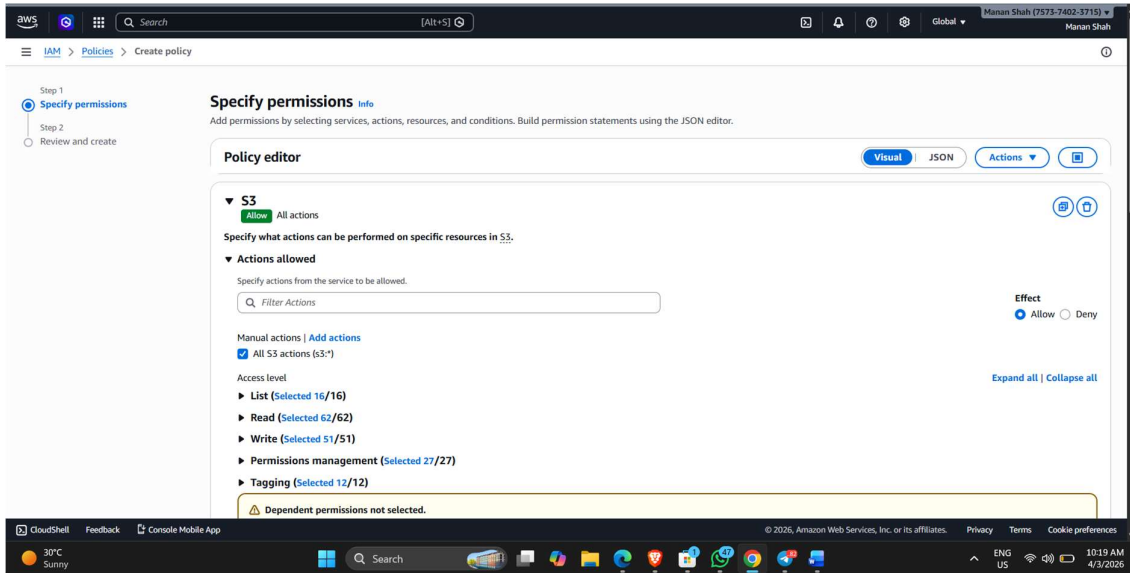
b) Set Up AWS Policy

Steps:

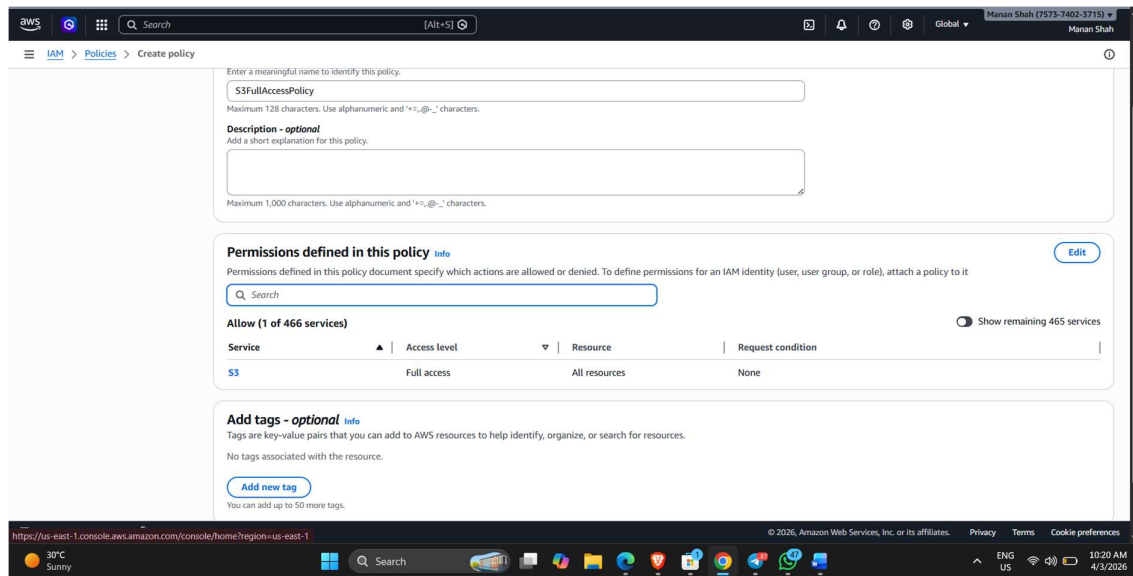
1. Go to IAM → Policies
2. Click Create Policy
3. Select Amazon S3 service
4. Allow all S3 actions
5. Select all resources
6. Create policy

Result:

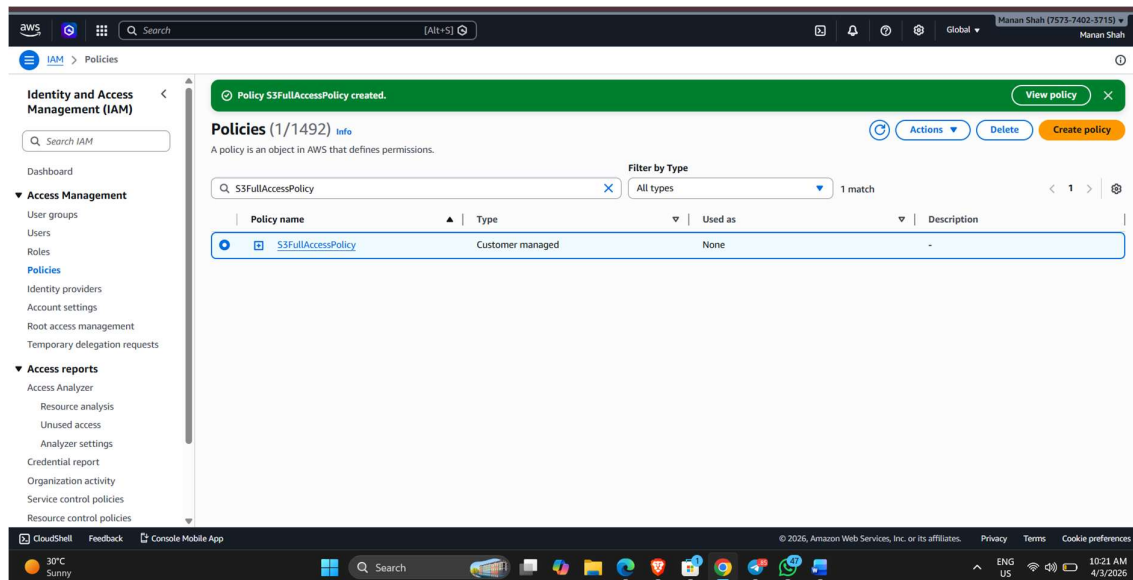
S3 access policy created successfully.



Click next



click create policy



Create IAM Role

Steps:

1. Go to IAM → Roles
2. Click Create Role
3. Select EC2 service
4. Attach S3FullAccessPolicy
5. Create role

Result:

IAM role created successfully and linked with EC2.

The image displays two screenshots of the AWS IAM console, illustrating the process of creating an IAM role for EC2.

Top Screenshot: Select trusted entity

The console shows the "Create role" wizard. The first step, "Select trusted entity", is active. The "Trusted entity type" section offers five options:

- AWS service** (selected): Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- AWS account**: Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- Web identity**: Allow users federated by the specified external web identity provider to assume this role to perform actions in this account.
- SAML 2.0 federation**: Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- Custom trust policy**: Create a custom trust policy to enable others to perform actions in this account.

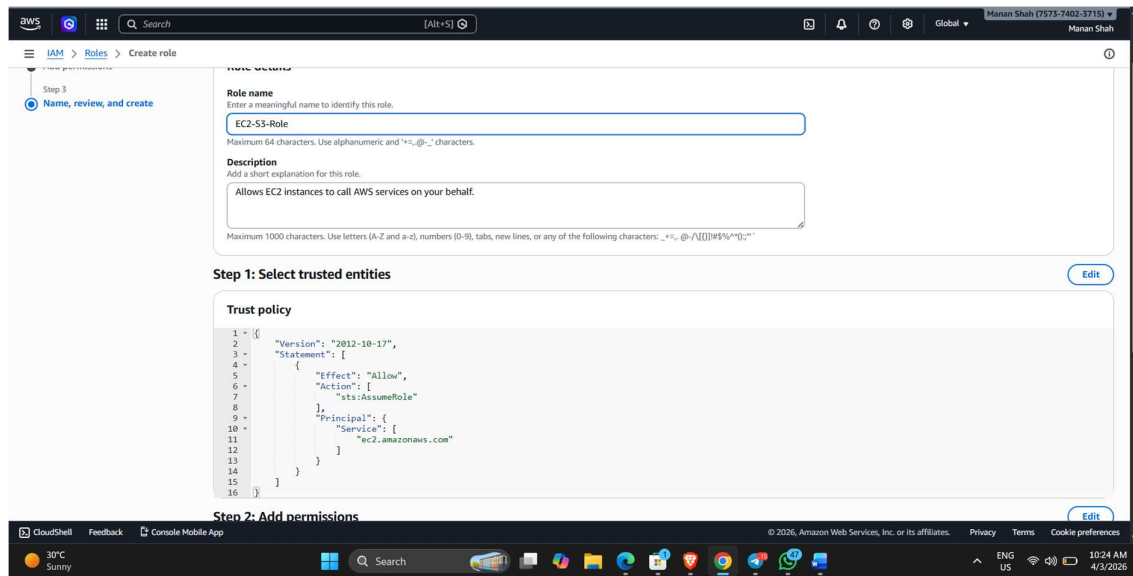
The "Use case" section is also visible, with "EC2" selected as the service or use case.

Bottom Screenshot: Add permissions

The second step, "Add permissions", is active. It shows "Permissions policies (1/1156)". A search bar contains "S3FullAccessPolicy", and the "Filter by Type" dropdown is set to "Customer managed", resulting in 1 match.

Policy name	Type	Description
☒ S3FullAccessPolicy	Customer managed	-

Below the table, there is a link to "Set permissions boundary - optional". At the bottom right, there are "Cancel", "Previous", and "Next" buttons.



Create Role

Attach IAM Role to EC2

Steps:

1. Go to EC2 → Instances
2. Select the instance
3. Click Actions → Security → Modify IAM Role
4. Select EC2-S3-Role
5. Click Update

Result:

IAM role successfully attached to EC2 instance.

EC2 > Instances

Instances (1/3) Info

Connect Instance state Actions Launch instances

Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
mywebserver	i-048a46708f10a482a	Running	t3.micro	3/3 checks passed	View	Public
mywebse	i-00d8b76a37d3acd85	Running	t3.micro	3/3 checks passed	View	52.90.1
web-server	i-0782204a3434aa8a2	Running	t3.micro	3/3 checks passed	View	54.91.1

Change security groups
Modify IAM role
Get Windows password

i-0782204a3434aa8a2 (web-server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID
i-0782204a3434aa8a2

Public IPv4 address
3.92.48.227 | open address

Private IPv4 addresses
172.31.88.97

IPv6 address
-

Instance state
Running

Public DNS
ec2-3-92-48-227.compute-1.amazonaws.com | open address

Hostname type
IP name: ip-172-31-88-97.ec2.internal

Private IP DNS name (IPv4 only)
ip-172-31-88-97.ec2.internal

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

30°C Sunny

Search

ENG US 10:25 AM 4/3/2026

EC2 > Instances > i-0782204a3434aa8a2 > Modify IAM role

Modify IAM role Info

Attach an IAM role to your instance.

Instance ID
i-0782204a3434aa8a2 (web-server)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

EC2-S3-Role

Create new IAM role

Cancel Update IAM role

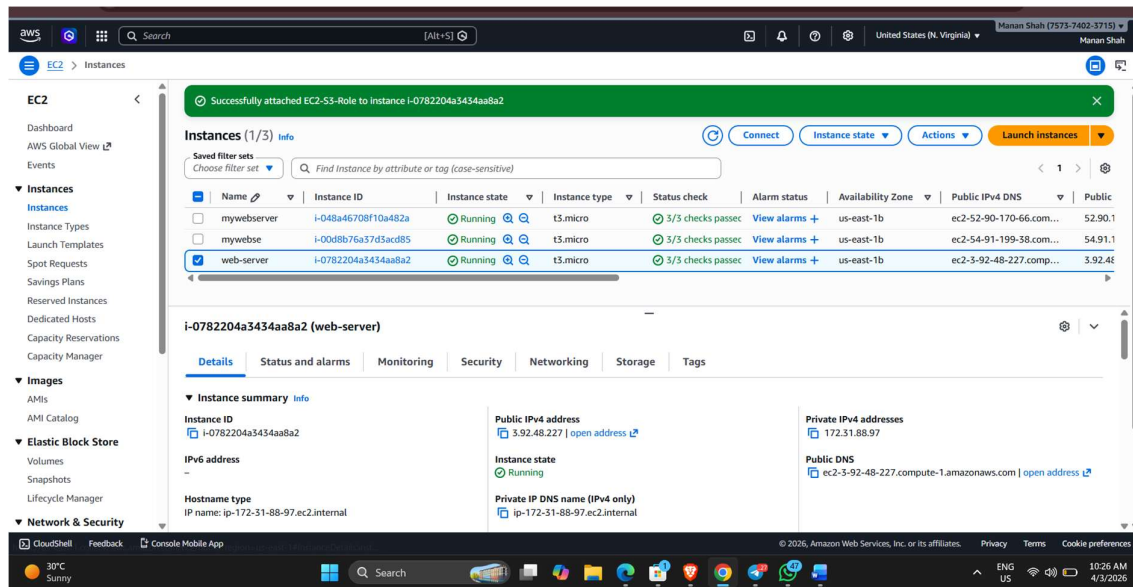
CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

30°C Sunny

Search

ENG US 10:26 AM 4/3/2026



Install and Configure AWS CLI

Steps:

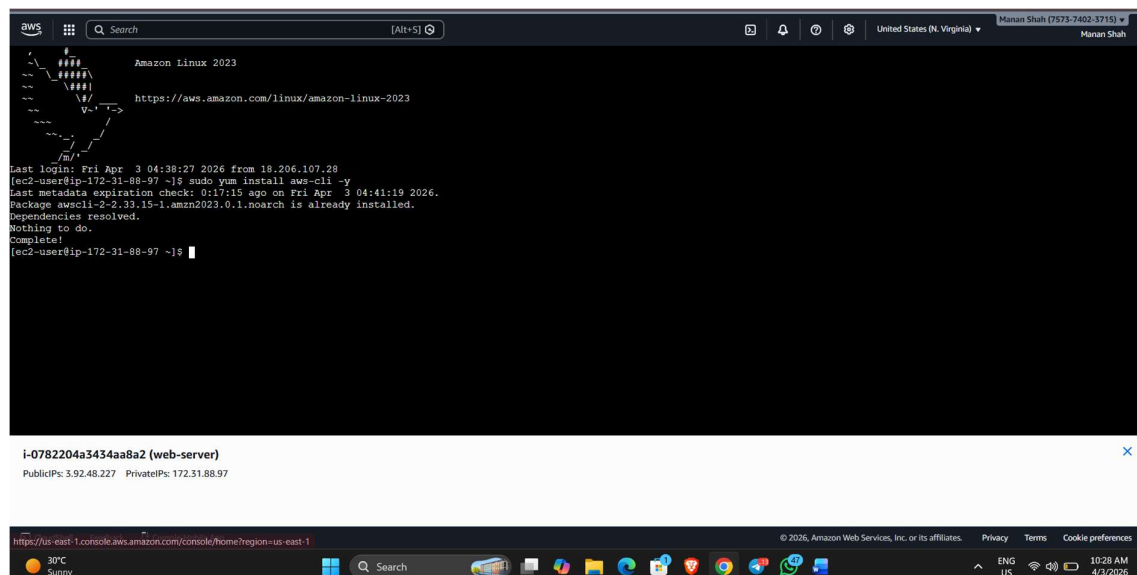
1. Connect to EC2 instance
2. Open terminal
3. Run command:

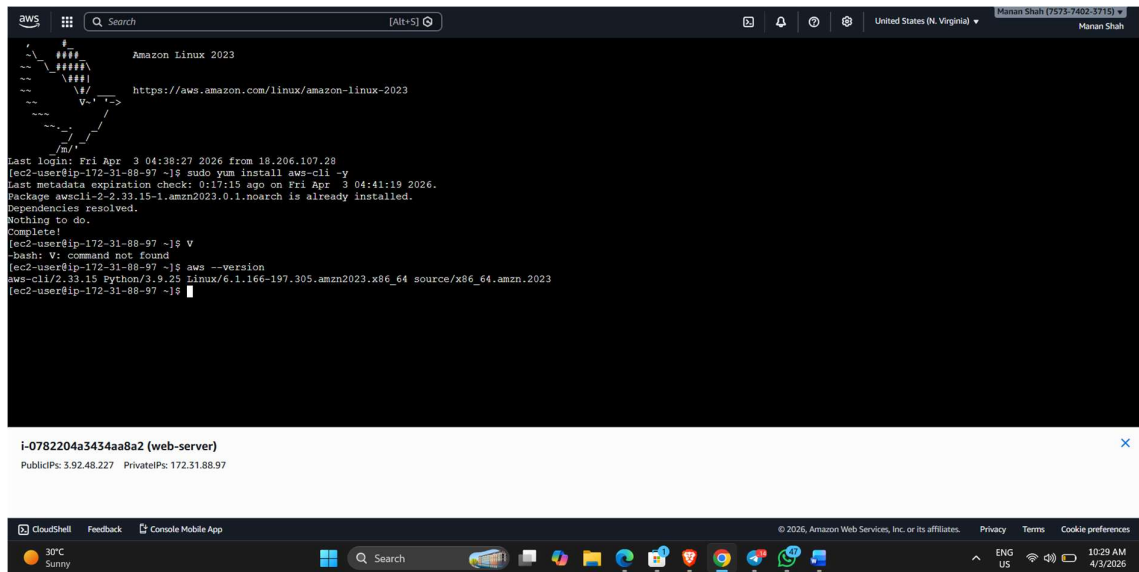
`sudo yum install aws-cli -y`

4. Run:

`aws configure`

5. Enter region: us-east-1





Create a Python Script

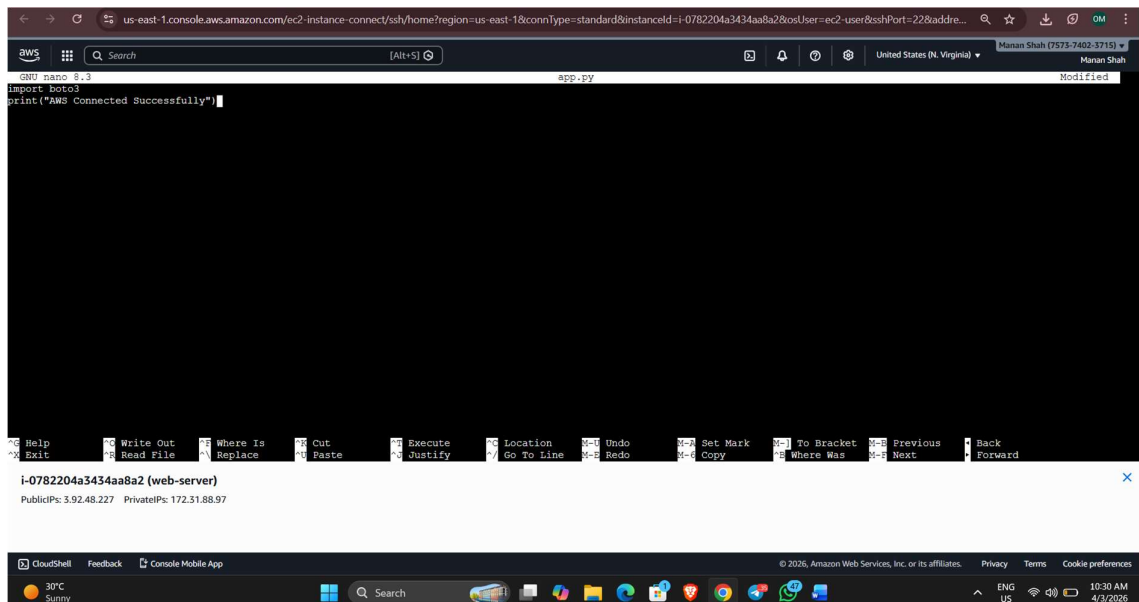
Steps:

1. Open terminal
2. Create file:

`nano app.py`

3. Write code:

```
import boto3  
print("AWS Connected Successfully")
```



Error Encountered

boto3 module not found and pip3 not installed

Solution

Installed pip and boto3 using commands

sudo yum install python3-pip -y

pip3 install boto3

```
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Installing : libxcrypt-compat-4.4.33-7.amzn2023.x86_64 1/2
  Installing : python3-pip-21.3.1-2.amzn2023.0.16.noarch 2/2
  Running scriptlet: python3-pip-21.3.1-2.amzn2023.0.16.noarch 2/2
  Verifying : libxcrypt-compat-4.4.33-7.amzn2023.x86_64 1/2
  Verifying : python3-pip-21.3.1-2.amzn2023.0.16.noarch 2/2

Installed:
  libxcrypt-compat-4.4.33-7.amzn2023.x86_64 python3-pip-21.3.1-2.amzn2023.0.16.noarch

Complete!
[ec2-user@ip-172-31-88-97 ~]$ pip3 --version
pip 21.3.1 from /usr/lib/python3.9/site-packages/pip (python 3.9)
[ec2-user@ip-172-31-88-97 ~]$ pip3 install boto3
Defaulting to user installation because normal site-packages is not writeable
Collecting boto3
  Downloading boto3-1.42.82-py3-none-any.whl (140 kB)
    |#####| 140 kB 17.6 MB/s
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/lib/python3.9/site-packages (from boto3) (0.10.0)
Collecting s3transfer<0.17.0,>=0.16.0
  Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
    |#####| 86 kB 9.8 MB/s
Collecting botocore<1.43.0,>=1.42.82
  Downloading botocore-1.42.82-py3-none-any.whl (14.8 MB)
    |#####| 14.8 MB 55.0 MB/s
Requirement already satisfied: urllib3<1.27,>=1.25.4 in /usr/lib/python3.9/site-packages (from botocore<1.43.0,>=1.42.82->boto3) (1.25.10)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/lib/python3.9/site-packages (from botocore<1.43.0,>=1.42.82->boto3) (2.8.1)
Requirement already satisfied: six>=1.5 in /usr/lib/python3.9/site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.43.0,>=1.42.82->boto3) (1.15.0)
Installing collected packages: botocore, s3transfer, boto3
Successfully installed boto3-1.42.82 botocore-1.42.82 s3transfer-0.16.0
[ec2-user@ip-172-31-88-97 ~]$

i-0782204a3434aa8a2 (web-server)
PublicIPs: 3.92.48.227 PrivateIPs: 172.31.88.97

CloudShell Feedback Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
30°C Sunny
```

NOW RUN

python3 app.py

```
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.2.x86_64 apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.66-1.amzn2023.0.1.x86_64 httpd-core-2.4.66-1.amzn2023.0.1.x86_64 httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-88-97 ~]$ python3 app.py
AWS Connected Successfully
[ec2-user@ip-172-31-88-97 ~]$

i-0782204a3434aa8a2 (web-server)
PublicIPs: 3.92.48.227 PrivateIPs: 172.31.88.97

CloudShell Feedback Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
30°C Sunny
```

The locally hosted web application was successfully migrated to AWS by deploying it on an EC2 instance. IAM roles and policies were configured, and AWS CLI along with a Python script was used to verify cloud connectivity. The successful output confirms that the cloud setup is working correctly.